Activity: Explore a Pre-Built Image Analysis Pipeline for Sensor Comparison

Introduction

In this activity, you'll explore a pre-built image analysis pipeline using OpenCV to compare images captured by **CMOS** and **CCD** sensors. You'll analyze the images for **noise** and **sharpness** using a standard pipeline, then modify the Canny edge detection thresholds to observe how this affects sharpness evaluation. This hands-on task introduces you to image analysis techniques while connecting sensor physics to practical outcomes for computer vision, all within a lightweight setup.

Learning Outcomes

By the end, you will:

1. **Use** a pre-built OpenCV pipeline to analyze CMOS and CCD sensor images.
2. **Modify** the Canny edge detection thresholds in the pipeline.
3. **Analyze** how the threshold change impacts sharpness evaluation.

Working with Dev Container

To complete this activity in a reliable and fully configured environment, please refer to the instructions in the file: `README-devcontainer.md`. This guide walks you through opening the assignment in **Visual Studio Code** using a **Dev Container**, which automatically installs all necessary Python libraries and tools. Following that setup ensures that the notebook runs smoothly without manual configuration or missing dependencies. Make sure to open the **main activity folder** in VS Code and follow the steps outlined in the Dev Container guide before starting the notebook.

Starter File

- Open the Jupyter notebook in the Code Folder (`activity-compare-sensor-types-and-image-quality.ipynb`) and follow the steps.
- Run on your local machine (CPU) or Colab (no GPU needed).

Requirements

- Analyze noise and sharpness for both CMOS and CCD images.
- Test two sets of Canny thresholds: default (100, 200) and modified (50, 150).
- Record noise levels and compare edge detection results.
- Plot edge detection results for both threshold settings.

Activity Code Steps

This is a guided learning activity. You'll:

1. **Load** and preprocess CMOS and CCD images by converting them to grayscale.
2. **Use** a pre-built pipeline to measure noise (standard deviation in a uniform patch) and sharpness (Canny edge detection).
3. **Analyze** noise and sharpness with default Canny thresholds (100, 200), recording noise levels and visualizing edges.
4. **Evaluate** the edge detection results for sharpness differences between CMOS and CCD.
5. **Modify** the Canny thresholds to (50, 150) and re-run the edge detection.
6. **Compare** sharpness evaluation between the two threshold settings using plots.

Conclusion

You'll have explored a lightweight image analysis pipeline, modified its Canny edge detection thresholds, and analyzed how this impacts sharpness evaluation for CMOS and CCD sensors. This prepares you for working with more advanced image processing techniques in computer vision, such as noise reduction or feature extraction.